

HOW LEARNING RATE DECAY WASTES YOUR BEST DATA IN CURRICULUM-BASED LLM PRETRAINING

Kairong Luo^{1*} Zhenbo Sun¹ Haodong Wen¹ Xinyu Shi¹
 Jiarui Cui¹ Chenyi Dang¹ Kaifeng Lyu^{1†} Wenguang Chen^{1,2†}

¹Tsinghua University

²Peng Cheng Laboratory

ABSTRACT

Due to the scarcity of high-quality data, large language models (LLMs) are often trained on mixtures of data with varying quality levels, even after sophisticated data curation. A natural approach to better leverage high-quality data is *curriculum-based pretraining*, where the model is trained on data sorted in ascending order of quality as determined by a quality metric. However, prior studies have reported limited improvements from such curriculum-based pretraining strategies. This work identifies a critical factor constraining these methods: the incompatibility between the ascending data quality order and the decaying learning rate (LR) schedule. We find that while curriculum-based training substantially outperforms random shuffling when using a constant LR, its advantage diminishes under standard LR decay schedules. Our experiments show this incompatibility can be mitigated by two simple strategies: (1) employing a more moderate LR decay schedule, where the final LR is only moderately smaller than the peak LR, and (2) replacing LR decay with model averaging, i.e., computing a weighted average of the final few checkpoints. By combining these strategies, we improve the average score on a suite of standard benchmarks by 1.64% over random shuffling, without additional data refinement. Validated on 1.5B-parameter models trained over 30B tokens with various data-quality metrics, our findings call for a re-evaluation of curriculum-based LLM pretraining and underscore the potential of co-designing data curricula with optimization methods.

1 INTRODUCTION

Large language models (LLMs) are typically trained on massive text corpora collected from the Internet (Dubey et al., 2024; DeepSeek-AI et al., 2024; Yang et al., 2025; OpenAI, 2023), covering a wide range of sources and quality levels. High-quality data plays a crucial role in enhancing model capabilities, but it is usually limited in amount. To address this issue, current LLM pre-training pipelines employ sophisticated data curation procedures to filter out low-quality data and increase the proportion of high-quality data, including rule-based (or heuristic-based) filtering, quality scoring (model-based labeling), and score-based data selection (Su et al., 2025; Li et al., 2024; Penedo et al., 2025; 2023; Weber et al., 2024). Despite these advances, relatively little attention has been given to developing *training strategies* that more effectively utilize the high-quality data during training, rather than only during data curation.

A natural idea to improve the utilization of high-quality data is to use *curriculum learning*¹. This is motivated by the catastrophic forgetting problem (McCloskey & Cohen, 1989), which refers to the phenomenon that a model may forget the knowledge it has learned before when it is exposed to new data (Dai et al., 2025; Liao et al., 2025). In contrast to random shuffling, this curriculum-based

*luokr24@mails.tsinghua.edu.cn

†Corresponding authors.

¹Traditionally, curriculum learning refers to training on progressively harder examples. Here, following prior work (Wettig et al., 2024), we generalize the term to denote data-ordering strategies such as progressing from low-quality to high-quality data.

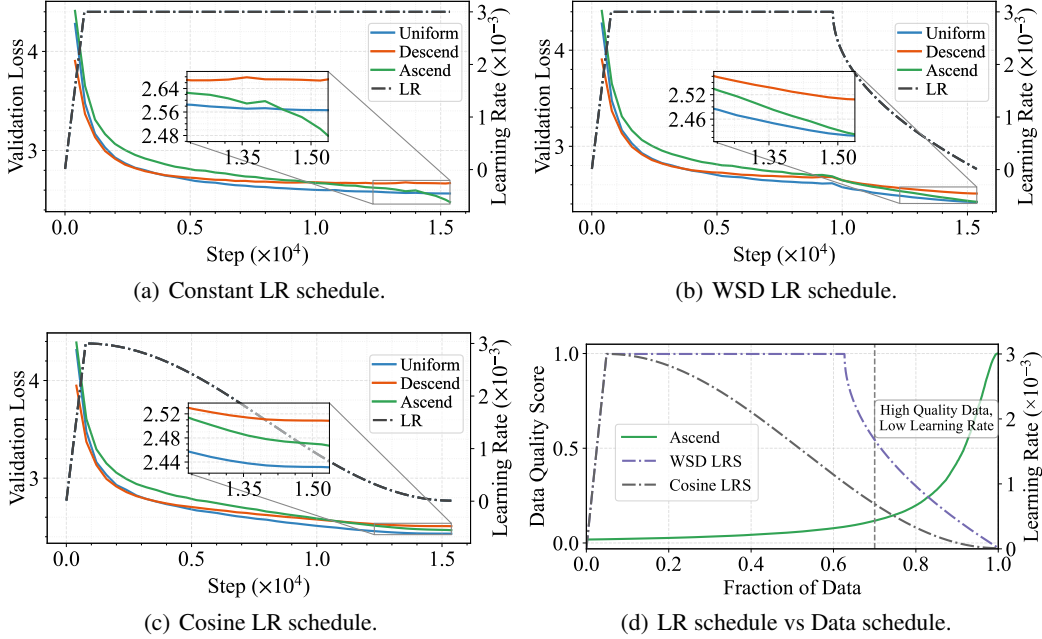


Figure 1: Data curriculum strategies are less effective when combined with learning rate (LR) schedules that decay to a low scale near the end. **(a-c)** Experiments on a 1.5B parameter model trained on 30B tokens compare various data curricula (Uniform, Ascending-Order, and Descending-Order by DCLM score (Li et al., 2024)) under constant, Warmup-Stable-Decay (WSD) (Hu et al., 2024; Hägele et al., 2024), and cosine schedules. While curricula improve validation loss over a uniform baseline with a constant LR, this advantage is significantly reduced during a low-LR phase following LR decay. **(d)** In the data curriculum, high-quality data is placed in the latter phase, which coincides with the LR decaying to a relatively low scale.

approach aims to optimize knowledge acquisition by exposing the model to high-quality data in the latter stages of training.

One successful curriculum learning strategy is multi-stage pretraining: first training on a data mixture dominated by massive web data, then in the second stage, referred to as *mid-training* (OLMo et al., 2025; Abdin et al., 2024b), shifting the data mixture to one that mainly consists of high-quality data. This strategy has been adopted by many recent LLMs, including OLMo 2 (OLMo et al., 2025), Phi-4 (Abdin et al., 2024a), and LongCat-Flash (Team et al., 2025). This two-phase design is most common, and it is also promising to extend with more stages (Yiwen et al., 2025; Allal et al., 2025) or follow with long-context extension (Yang et al., 2025).

Another line of work explores curriculum learning at the *instance level*, where data samples are sorted according to quality scores and presented to the model sequentially (Wettig et al., 2024; Dai et al., 2025; Zhang et al., 2025; Kim & Lee, 2024). We refer to this as the *data curriculum*². However, these studies mainly investigate different quality metrics and find that simple end-to-end sorting yields limited benefits. Consequently, several works propose alternative strategies such as *folding curriculum* (Detailed in Section 4), which reorders samples within consecutive phases in an interleaved manner (Dai et al., 2025; Zhang et al., 2025). Despite showing promise, we find that this interleaved approach is fragile: its advantage does not extend to our larger-scale experiments with the DCLM fastText score (Li et al., 2024), a widely used scoring metric (see Section 4).

This raises a central question: *Why do instance-level curriculum learning strategies often yield limited benefits?* This is not simply due to unreliable quality scores: metrics like the QuRating score (proposed and used by Wettig et al. (2024)), the PDS score (proposed by Gu et al. (2025b) and discussed by Dai et al. (2025)), and the DCLM score (Li et al., 2024) are already informative enough to improve training efficiency by guiding high-quality data selection.

Our Contributions. In this paper, we identify a key, yet previously overlooked factor: *the incompatibility between the ascending order of data quality and the decaying schedule of learning rate*. As illustrated in Figure 1, if we train an LLM with a constant LR, using a data curriculum that sorts

²We use *data schedule* as a general term for any strategy that specifies the order of training data. Unless stated otherwise, *data curriculum* denotes a schedule that sorts data samples in ascending or descending order (reverse curriculum) with respect to a particular quality metric.

data in ascending order of quality can indeed outperform the baseline that trains the model on data in a uniform order. However, when we switch to a more standard LR decay schedule, such as cosine or Warmup-Stable-Decay (WSD) (Loshchilov & Hutter, 2017; Hu et al., 2024) (a schedule with warmup, plateau, and decay phases, see Figure 1(b)), the benefit of the data curriculum diminishes. Moreover, we observe that as the LR decay becomes more aggressive (e.g., having a longer decay phase or a lower ending LR), the benefit of the data curriculum diminishes more.

To resolve the incompatibility between data curriculum and LR decay, firstly, we discuss a straightforward remedy: adopting a moderate LR decay schedule in curriculum learning. In Section 5.1, we show that tuning the ending LR in WSD schedules trades off the benefits of data curriculum and loss convergence, and setting it to a moderate value (approximately decaying to 1/3 of peak LR in our setting) can make use of high-quality data and outperform uniform data ordering.

Further, we propose a strategy that largely resolves the incompatibility between the data curriculum and the LR schedule by replacing LR decay with *weight averaging* (Li et al., 2025c; Izmailov et al., 2018; Tian et al., 2025). Weight averaging computes a weighted average of recent checkpoints as the final model. It stabilizes model parameters and reduces noise in the training process, similar to the effect of LR decay. However, it can achieve this without diminishing the magnitude of updates. Therefore, we adopt a constant LR throughout training and pair weight averaging with the data curriculum. This combination allows the model to maintain a high learning rate and fully exploit the high-quality data introduced later in the curriculum. We call this approach *Curriculum Model Averaging (CMA)*. Notably, CMA also extends emerging practices of introducing weight averaging into LLM training (Tian et al., 2025; Li et al., 2025c; Sanyal et al., 2024), showing that combining weight averaging with curriculum learning yields greater benefits than applying weight averaging to standard uniform-ordering pretraining. This combination is particularly effective under multi-phase pretraining, where high-quality data is introduced in the mid-training phase. In this setting, our approach achieves an average improvement of 1.2% in accuracy—and over 2% on core benchmarks (defined in Section 4)—solely by reordering data samples (Table 2).

Building on these explorations, we further demonstrate that combining moderate LR decay, curriculum learning, and weight averaging can produce synergistic advantages and reveal a previously overlooked high-performing pretraining regime, improving standard benchmarks by 1.64% in average accuracy over random shuffling and standard decay. This finding underlines that hyperparameter settings optimized for uniform data ordering are not necessarily optimal for curriculum-based training. Prior work has naturally evaluated curriculum-based training using regimes originally optimized for uniform data ordering. However, this regime is suboptimal for curriculum-based training and, as previously reported, leads to only marginal benefits (Wettig et al., 2024; Kim & Lee, 2024). In contrast, we identify a previously underexplored and more effective regime in the design space of LLM pretraining by co-designing these components, paving the way for future exploration.

We validate our hypotheses and proposed strategies through experiments at a scale sufficient to support our conclusions, training a 1.5 billion-parameter model on 30 billion tokens. Our findings demonstrate robustness across different quality metrics, LR schedules, and data mixtures, highlighting that co-designing data curricula with training dynamics is a powerful, data-aware strategy for improving LLM pretraining efficiency.

2 RELATED WORK

Curriculum Learning in LLM Pretraining. While data quality is known to be critical, leveraging it through fine-grained, instance-level curricula has shown limited success and has not been validated at a substantial scale (Dai et al., 2025; Zhang et al., 2025; Wettig et al., 2024; Campos, 2021). Prior studies have reported only marginal gains, sometimes observing benefits from counter-intuitive data orders without a clear underlying mechanism (Wettig et al., 2024). Other works either lack sufficient validation (Campos, 2021; Kim & Lee, 2024) or, finding simple curricula ineffective, propose more complex strategies like data folding (Zhang et al., 2025; Dai et al., 2025). However, we find that the benefits of such complex strategies are often confined to smaller-scale experiments with low learning rates and can degrade at larger scales and under a high LR regime (Section B.1). A common thread in these attempts is the use of standard cosine learning rate schedules. We demonstrate that decayed LR prevents the model from effectively learning from the high-quality data introduced late in the training process, showing that a more moderate decay is required to unlock the curriculum’s full potential. A more detailed discussion can refer to Section B.1.